

enabling end-to-end learning without manual feature engineering.

The proposed approach is evaluated using the widely adopted RadioML2016.10a dataset, which contains multiple analog and digital modulation schemes across a wide range of SNR conditions. Experimental results demonstrate that the residual architecture improves learning stability and classification accuracy, particularly under moderate and high SNR levels.

The main contributions of this work are summarized as follows:

1. A Residual 1D CNN architecture designed for automatic modulation classification from raw IQ signals.
2. End-to-end deep learning framework eliminating handcrafted feature extraction.
3. Improved convergence behavior through residual learning mechanisms.
4. Comprehensive performance evaluation across varying SNR levels using the RadioML2016.10a dataset.

II. RELATED WORK

Automatic Modulation Classification (AMC) has attracted significant research attention due to its essential role in intelligent wireless communication systems, spectrum monitoring, and cognitive radio applications. Over time, AMC techniques have evolved from traditional statistical approaches to modern deep learning-based solutions capable of learning directly from raw signal representations.

performance under ideal channel assumptions; however, they require accurate prior knowledge of channel parameters such as noise variance and synchronization information. Furthermore, their computational complexity grows rapidly with increasing modulation classes, making them impractical for real-time wireless environments.

To address these limitations, feature-based machine learning methods were introduced. These approaches extract handcrafted features from received signals, including higher-order statistics, cyclic spectral features, and instantaneous amplitude or phase characteristics, followed by classification using algorithms such as Support Vector Machines (SVM), decision trees, or k-Nearest Neighbors (k-NN). Although these methods reduce computational cost compared to likelihood-based approaches, their performance strongly depends on expert-designed features and often degrades under low Signal-to-Noise Ratio (SNR) conditions and varying channel environments.

Recent advances in deep learning have significantly transformed AMC research by enabling automatic feature extraction directly from raw in-phase and quadrature (IQ) signal samples. Convolutional Neural Networks (CNNs) have demonstrated strong performance due to their ability to learn hierarchical temporal representations without manual feature engineering. Several studies have shown that CNN-based models outperform traditional approaches, particularly in noisy scenarios where handcrafted features become unreliable.

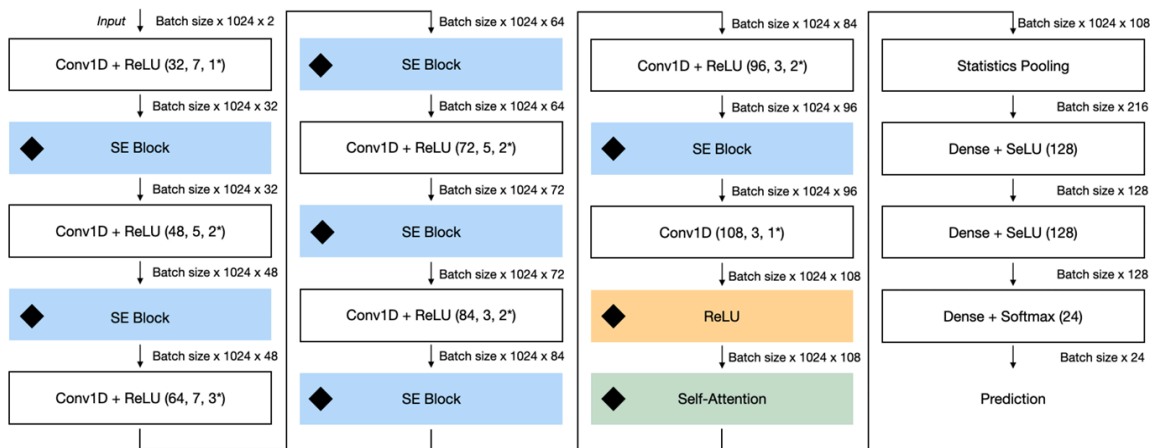


Fig 1. Proposed architecture by Harper et al [1].

Early AMC methods primarily relied on likelihood-based techniques, which formulate modulation recognition as a statistical hypothesis testing problem. These approaches achieve optimal

Beyond conventional CNN architectures, researchers have investigated deeper and more specialized neural network designs to improve classification performance. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models have

been explored to capture temporal dependencies in communication signals; however, these architectures often introduce increased computational complexity and longer training times. Hybrid CNN–RNN models have also been proposed, but their architectural complexity can lead to training instability.

A comprehensive architectural investigation was conducted by Harper *et al.* [1], who analyzed multiple deep neural network configurations for automatic modulation classification. Their study demonstrated that architectural components such as dilated convolutions, statistics pooling, and squeeze-and-excitation mechanisms significantly influence classification performance.

The authors showed that effective feature aggregation and channel attention mechanisms improve robustness across modulation types and SNR conditions, highlighting the importance of network design choices rather than merely increasing model depth.

Motivated by these findings, recent research has explored alternative deep learning architectures that improve training stability and feature learning efficiency. Residual learning frameworks introduce shortcut connections that facilitate gradient propagation and mitigate vanishing gradient problems in deep networks. Residual networks enable deeper hierarchical feature extraction while maintaining stable convergence behavior, making them well suited for time-series signal classification tasks.

Inspired by advances in deep neural architectures for AMC, this work adopts a Residual 1D Convolutional Neural Network designed to learn discriminative representations directly from raw IQ signals. The proposed approach focuses on improving convergence stability and classification robustness while maintaining computational efficiency for practical wireless signal classification systems.

III. DATASET AND PREPROCESSING

A. Dataset Description

The performance of the proposed Automatic Modulation Classification (AMC) model is evaluated using the widely adopted **RadioML2016.10a** dataset, which has become a benchmark dataset for deep learning-based modulation recognition research. The dataset contains synthetically generated radio communication signals representing multiple analog

and digital modulation schemes under varying channel conditions.

The RadioML2016.10a dataset includes **11 modulation classes**, comprising both digital and analog modulation types: BPSK, QPSK, 8PSK, CPFSK, GFSK, PAM4, QAM16, QAM64, AM-DSB, AM-SSB, and WBFM. Each signal sample consists of **128 complex baseband samples**, represented by in-phase (I) and quadrature (Q) components. These IQ samples are stored as two real-valued channels, forming an input tensor of dimension $128 \times 2128 \times 2$.

Signals are generated across a wide range of **Signal-to-Noise Ratio (SNR)** values from -20 dB to 18 dB with a step size of 2 dB, enabling evaluation of model robustness under diverse noise conditions. The dataset contains approximately **220,000 labeled signal samples**, evenly distributed across modulation categories and SNR levels.

B. IQ Signal Representation

In modern wireless communication systems, transmitted signals are commonly represented using complex baseband notation. Each signal sample is composed of an in-phase component $I(t)$ and a quadrature component $Q(t)$, which together encode amplitude and phase information of the waveform. Instead of extracting handcrafted features, the proposed approach directly utilizes raw IQ samples as input to the neural network, allowing automatic feature learning through convolutional operations.

The IQ representation preserves temporal and phase relationships within the signal, enabling convolutional neural networks to learn discriminative modulation patterns directly from time-domain data.

C. Data Preprocessing

Prior to training, several preprocessing steps are applied to ensure stable learning and improved convergence of the neural network model.

1) Data Reshaping

The original dataset stores signals in the format $(2, 128)$, corresponding to IQ channels and temporal samples. For compatibility with one-dimensional convolutional layers, the data is transposed into the format:

$(\text{samples}, 128, 2)$

where 128 represents the temporal dimension and 2 corresponds to the IQ channels.

2) Normalization

To reduce amplitude variations and stabilize gradient updates during training, each signal sample is normalized individually using maximum absolute scaling:

$$X_{norm} = \frac{X}{\max(|X|)}$$

This per-sample normalization ensures that signal values lie within a consistent numerical range while preserving relative waveform characteristics.

3) Label Encoding

Modulation labels are converted into numerical class indices and subsequently transformed into one-hot encoded vectors for multi-class classification. This representation enables the use of categorical cross-entropy loss during model training.

4) Train–Test Split

The dataset is divided into training and testing subsets using a stratified sampling strategy to preserve class distribution across modulation types. Approximately 80% of the data is used for training, while the remaining 20% is reserved for performance evaluation. Stratification ensures balanced representation of all modulation categories during learning and testing phases.

D. Evaluation Across SNR Levels

Since modulation classification performance strongly depends on channel noise conditions, evaluation is conducted across multiple SNR values. This allows analysis of model robustness under both low-SNR (noisy) and high-SNR (clean signal) environments, providing a comprehensive assessment of classification capability.

IV. PROPOSED METHODOLOGY

A. Overview of the Proposed Framework

This work proposes a Residual 1D Convolutional Neural Network (ResNet-1D) architecture for automatic modulation classification directly from raw IQ signal samples. The model is designed to capture temporal signal characteristics while maintaining stable training behavior through residual learning. Classification using a fully connected layer with a softmax activation function.

Residual learning enables deeper networks to be trained effectively by allowing shortcut connections that propagate gradients more efficiently during backpropagation. This improves convergence speed and prevents degradation problems commonly encountered in deep neural networks.

B. Residual Convolutional Architecture

The proposed architecture processes input signals of dimension (128×2) , where 128 represents the temporal samples and 2 represents the in-phase and quadrature components.

The network begins with an initial **1D convolutional layer** that extracts low-level temporal features from the input signal. This layer is followed by **Batch Normalization** and **ReLU activation**, which improve training stability and introduce non-linearity into the model.

After the initial feature extraction stage, the network employs a sequence of **residual blocks** with increasing numbers of convolutional filters. Each residual block contains two convolutional layers along with a shortcut connection that directly adds the input of the block to its output. This skip connection allows the network to learn residual mappings rather than complete transformations, facilitating deeper feature learning without suffering from vanishing gradient issues.

The architecture uses three residual stages with **64, 128, and 256 convolutional filters**, respectively. Between residual stages, **max-pooling layers** are applied to progressively reduce the temporal resolution while increasing the feature dimensionality. This hierarchical representation enables the network to capture both local signal structures and broader

The overall system framework is illustrated in **Fig. 2**. The input IQ signals are first preprocessed and normalized before being fed into a deep convolutional network composed of residual blocks. The network progressively extracts hierarchical features from the temporal signal representation, followed by global feature aggregation and temporal dependencies.

C. Feature Aggregation

Following the convolutional stages, a **Global Average Pooling (GAP)** layer is applied to aggregate the learned feature maps into a compact feature vector. Global pooling reduces the total number of trainable parameters and helps prevent

overfitting by eliminating fully connected layers with large parameter counts. The aggregated features are then passed through a **dense layer containing 256**

The learning rate is dynamically adjusted during training using a learning rate scheduler to improve convergence and avoid local minima.

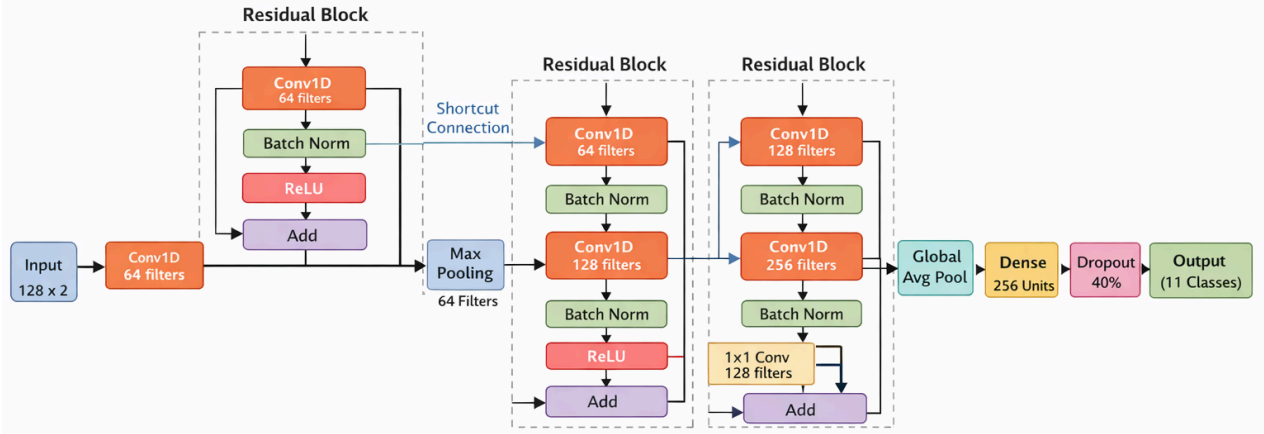


Fig. 2. Proposed Residual 1D CNN architecture for automatic modulation classification using raw IQ signals.

neurons with ReLU activation. A **dropout layer** is incorporated after the dense layer to improve model generalization by randomly deactivating neurons during training.

D. Output Layer and Classification

The final classification stage uses a **softmax output layer** that produces probability scores across the modulation classes. Each output node corresponds to a specific modulation type in the dataset.

The predicted modulation class is determined by selecting the class with the highest probability value. The network is trained using **categorical cross-entropy loss**, which measures the difference between predicted class probabilities and the ground truth labels.

E. Training Strategy

The model is trained using mini-batches of signal samples to improve computational efficiency. Key training parameters used in the experiments include:

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Batch Size	512
Epochs	100
Loss Function	Categorical cross entropy

V. EXPERIMENTAL RESULTS AND ANALYSIS

A. Experimental Setup

To evaluate the effectiveness of the proposed Residual 1D Convolutional Neural Network, experiments were conducted using the **RadioML2016.10a dataset** described in Section III. The dataset includes multiple analog and digital modulation schemes across varying Signal-to-Noise Ratio (SNR) levels ranging from -20 dB to 18 dB.

The model was implemented using the **TensorFlow deep learning framework** and trained on a system equipped with an **NVIDIA RTX 3050 GPU**. Training was performed using mini-batch gradient descent with the Adam optimizer.

The dataset was divided into training and testing subsets using an 80–20 split while maintaining balanced class distribution across modulation types. The model was trained for 30 epochs with a batch size of 512 samples.

B. Training Performance

The training performance of the proposed model was evaluated by monitoring both **training and validation accuracy and loss values** during the learning process. The convergence behavior of the model is illustrated in Fig. 3.

As shown in the figure, the training accuracy steadily increases over successive epochs while the loss function decreases, indicating effective learning of signal features. The validation accuracy follows a similar trend, demonstrating good generalization performance and minimal overfitting.

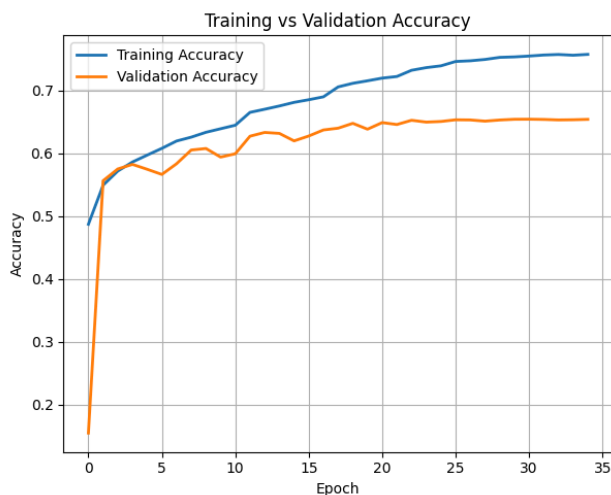


Fig 3(a). Training vs Validation Accuracy

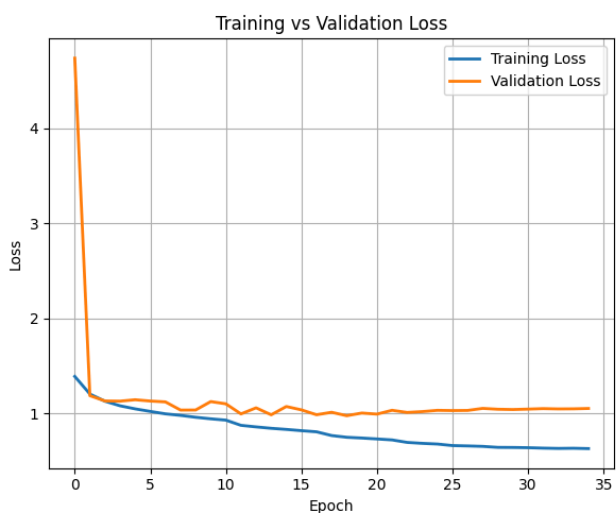
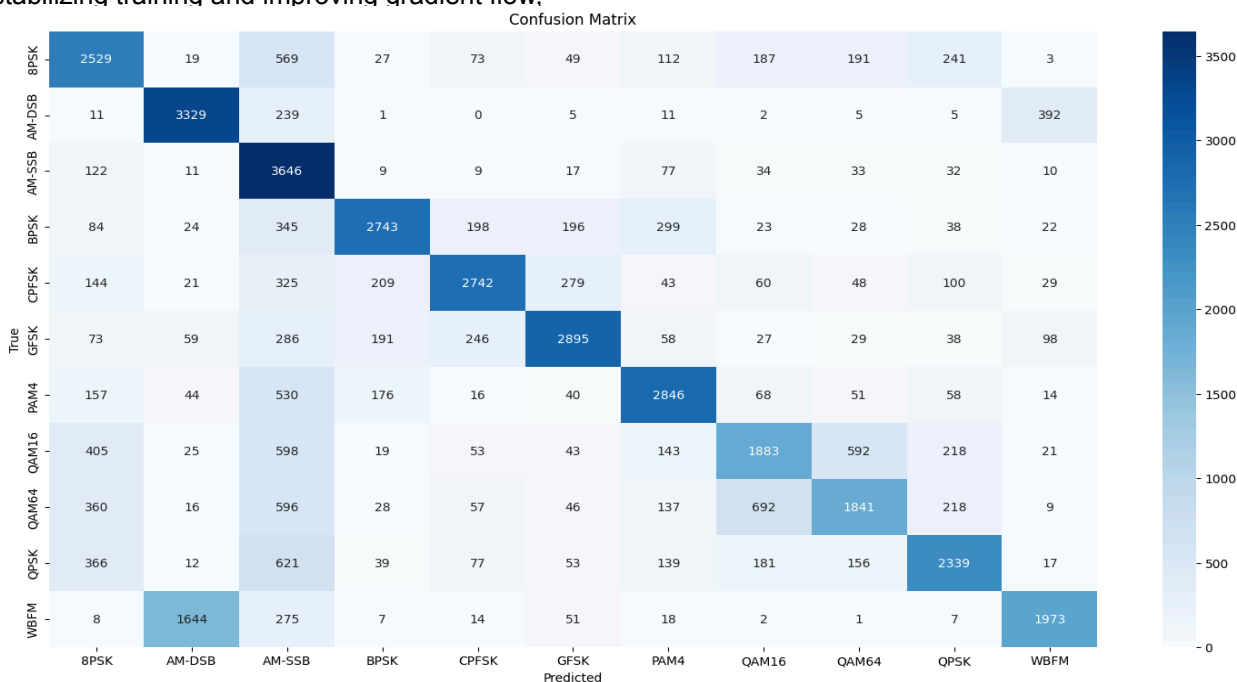


Fig 3(b). Training vs Validation Loss

Residual connections play an important role in stabilizing training and improving gradient flow,



allowing the network to learn deeper feature representations from the IQ signals.

C. Confusion Matrix Analysis

To further evaluate the classification performance of the model across different modulation schemes, a **confusion matrix** was generated using the test dataset. The confusion matrix is shown in **Fig. 4**, where rows represent true modulation labels and columns represent predicted labels. The confusion matrix reveals that the model correctly classifies the majority of modulation classes with high accuracy. However, certain modulation types with similar constellation characteristics, such as **QAM16** and **QAM64**, exhibit occasional misclassification, particularly under lower SNR conditions.

Overall, the confusion matrix demonstrates that the proposed residual architecture effectively distinguishes between both analog and digital modulation schemes.

D. Classification Accuracy Across SNR Levels

Since wireless communication environments are highly influenced by noise, it is essential to evaluate model performance across different Signal-to-Noise Ratio (SNR) levels. The classification accuracy of the proposed model was measured for each SNR value in the dataset, and the results are illustrated in Fig. 5.

As expected, the model achieves lower accuracy at extremely low SNR values due to heavy noise interference. However, as the SNR increases, the accuracy improves significantly because the modulation patterns become more distinguishable.

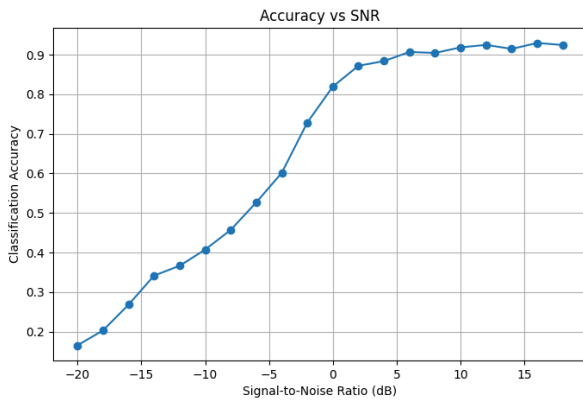


Fig. 5. Accuracy versus Signal-to-Noise Ratio (SNR) demonstrating the robustness of the proposed Residual 1D CNN model across different channel noise levels.

E. Discussion

The experimental results demonstrate that the proposed Residual 1D CNN architecture effectively learns discriminative features directly from raw IQ signals. The overall accuracy of the model is 65.38% compared to conventional CNN architectures, the inclusion of residual connections improves training stability and enables deeper feature extraction.

Per-Class Accuracy:

Class	Precision	Recall	f1-Score	Support
8PSK	0.59	0.63	0.61	4000
AM-DSB	0.64	0.83	0.72	4000
AM-SSB	0.45	0.91	0.61	4000
BPSK	0.80	0.69	0.74	4000
CPFSK	0.79	0.69	0.73	4000
GFSK	0.79	0.72	0.75	4000
PAM4	0.73	0.71	0.72	4000
QAM16	0.60	0.47	0.53	4000
QAM64	0.62	0.46	0.53	4000
QPSK	0.71	0.58	0.64	4000
WBFM	0.76	0.49	0.60	4000

The model performs particularly well at moderate and high SNR levels, where modulation patterns become clearer. Minor confusion between modulation types with similar constellation structures is expected and at moderate to high SNR levels, the model achieves strong classification performance, consistent with prior AMC studies.

Overall, the results confirm that deep residual learning provides a powerful approach for automatic modulation classification in modern wireless communication systems.

F. Comparison with Existing Work

To evaluate the effectiveness of the proposed Residual 1D CNN model, a comparative analysis is performed with the deep neural network architecture proposed by Harper *et al.* [1]. Their study emphasized the impact of architectural components such as dilated convolutions, statistics pooling, and squeeze-and-excitation (SE) blocks on modulation classification performance.

The proposed model differs in design by utilizing residual learning mechanisms to improve gradient propagation and training stability while maintaining a relatively simpler architecture.

This result indicates that the proposed deep residual architecture is capable of extracting robust signal features even under noisy channel conditions.

Feature	Herper et al.[1]	Proposed Model
Architecture	Dilated CNN + SE + Stats Pooling	Residual 1D CNN
Complexity	High	Moderate
Feature Extraction	Advanced(attention + pooling)	Residual Feature Learning
Training Stability	Good	Improved (via residual connections)
Overall Accuracy	63.7%	65.38%
High SNR Accuracy	High	High
Implementation Complexity	Complex	Simpler

VI. CONCLUSION

This paper presented a deep learning-based approach for **Automatic Modulation Classification (AMC)** using a **Residual 1D Convolutional Neural Network** applied directly to raw IQ signal samples. The proposed architecture leverages residual

connections and hierarchical convolutional feature extraction to improve learning stability and enable efficient representation of temporal signal characteristics. By eliminating the need for handcrafted feature engineering, the model is able to automatically learn discriminative features from the received signals.

The proposed method was evaluated using the **RadioML2016.10a dataset**, which contains multiple analog and digital modulation schemes across a wide range of Signal-to-Noise Ratio (SNR) conditions. Experimental results demonstrate that the residual CNN architecture achieves strong classification performance and stable convergence during training. Analysis of the confusion matrix indicates that the model effectively distinguishes between most modulation types, while minor misclassifications occur primarily among modulation schemes with similar constellation structures. Furthermore, performance evaluation across SNR levels shows that classification accuracy improves significantly as the signal quality increases.

Overall, the results confirm that deep residual learning provides an effective solution for automatic modulation recognition in modern wireless communication systems. Future work may explore advanced architectures such as attention mechanisms, transformer-based models, or hybrid deep learning frameworks to further enhance classification accuracy and robustness under challenging channel conditions.

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